

LArASIC P5 Datasheet

Dr. Venkata Narasimha Manyam

Dr. Giovanni Pinaroli

Dr. Grzegorz Deptuch

Revision Number	Change Description	Author	Revision Date
1.0	Initial draft based on LArASIC P4 datasheet and with new modifications	VNM	Apr, 2021

LArASIC

LArASIC is a programmable, full-functionality 16-channel front-end ASIC in 180 nm CMOS designed for the Deep Underground Neutrino Experiment (DUNE) single-phase (SP) liquid argon (LAr) time-projection chamber (TPC). It is capable of performing a low-noise, highly-linear charge amplification by a factor of up to 320x, higher-order shaping with a charge to voltage conversion, ac/dc coupling along with high-performance buffering in each channel. Signals can be read out using direct outputs of the shaping amplifiers, single-ended output buffers, or single-to-differential converting output buffers. It is designed and optimized to operate at the cryogenic temperature of 87 K, reliably for over 20 years of the experiment's lifetime. A simplified block diagram is as shown in Fig. 1

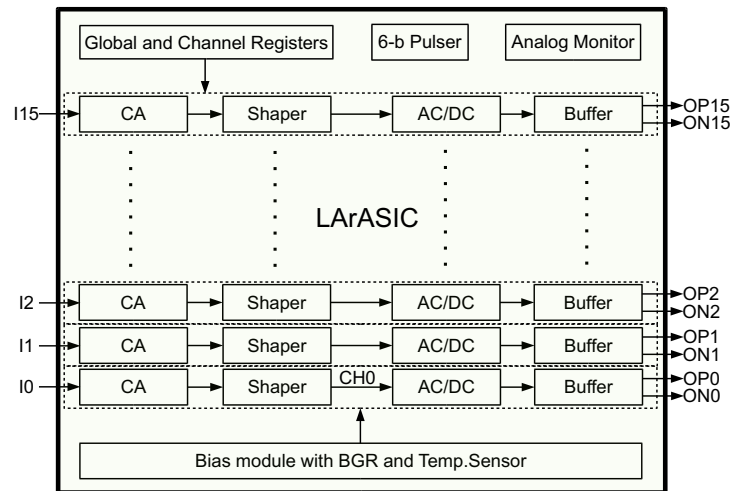


Fig. 1. Simplified block diagram of LArASIC

The ASIC is programmable with a total of 1024 different configurations possible for each channel's gain, peaking time, baseline selection, adaptive-reset quiescent currents, ac/dc coupling and single-ended (SE) or single-ended-to-differential conversion (SEDC) buffer enable. With I0~I15 inputs the ASIC makes the analog amplitudes available at the pins OP0~OP15 with SEDC disabled and OP0~OP15 as well as ON0~ON15 when SEDC is enabled. When the SEDC is selected, the outputs ON0~N15 are set in the high impedance state. The ASIC has in total 144 configuration register bits, 2 global and 16 channel registers each 8 bit long. All the required bias voltages are generated internally, on-chip using BJT bandgap reference. The output of the shaper or buffers (OP for SEDC) or temperature can be observed through the analog monitor output.

A simplified block diagram of the LArASIC channel is as shown in Fig. 2. The charge input to the LARASIC is amplified with the help of a dual-stage charge amplifier, with a gain of 20 and the selectable gain from the {3/5/9/16} set for the 1st and 2nd stage, respectively.

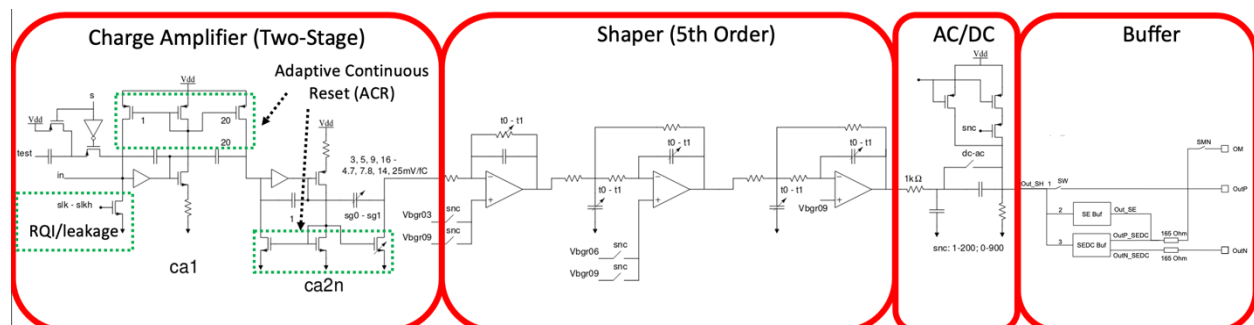


Fig. 2. A simplified block diagram of LArASIC channel

A 5th order shaping amplifier with complex conjugate poles converts the charge to voltage and performs shaping to maximize the SNR. The output of the shaper can be accessed with DC or AC coupling with 100 μ s HPF time-constant. The buffer block consists of selectable single-ended (SE) or single-ended-to-differential conversion (SEDC) buffer capable of driving subsequent COLDADC stage.

A brief synopsis along with specifications of the LArASIC are summarized in Table 1. The power consumption of the various subsystems of LArASIC is summarized in Table 2.

Table 1. LArASIC specifications and functionality

Technology	180 nm CMOS – 1-poly, 6-metal, MiM cap, silicide block resistors			
Supply Voltage	1.8 V			
Temperature Range	77 – 300 K (-196 – 27 °C) optimized for 87k (-186 °C)			
Max Single-Ended Output Swing	1.4 V peak to peak (low level 0.2 – high level 1.6 V)			
Gain Selection (mV/fC)	4.7	7.8	14	25
Full-Scale Input Charge (fC)	300	180	100	56
Baseline selection	200 mV (collection mode)		900 mV (induction/non-collection mode)	
Charge Preamplifier Polarity	Negative (collection mode)		Both (induction/non-collection mode)	
Adaptive-Reset Current Selection (nA)	0.1	0.5	1	5
Shaper Peaking Time Selection (μ s) given as 5% rise to peak time	0.5	1	2	3
Output Coupling	AC (100 μ s HPF time-constant)		DC	
Output Selection	Shaper		SE buffer	SEDC buffer
Total Channel Settings	1024			
Integrated Test Charge Injection Cap.	200 fF			
Temperature Sensor	0.8728 V @ 25°C + 2.868 mV/°C			
Integrated Pulse Generator	6-bit DAC based (with optional slope matching with Ch0 gain)			
Configuration Control	SPI interface with 144 register bits			

Table 2. Summary of LArASIC power consumption

Channels	~ 6 mW/ch without buffer ~ 10 mW/ch with single-ended buffer ~ 12 mW/ch with differential buffer	
Channel subsystems	Input MOSFET	~ 3.9 mW
	Charge Amp.	~ 0.8 mW
	Shaper	~ 1 mW
	SE buffer	~ 4 mW
	SEDC buffer	~ 6 mW
Common Circuitry	~ 4 mW	
Total Power Consumption	~ 100 mW without buffer	
	~ 164 mW with single-ended buffer	
	~ 196 mW with differential buffer	

Updates to this version:

- RQI subtraction in CA2n disconnected for guaranteed functionality at cold
- SEDC stability and CM noise rejection improved with modifications to the CMFB circuit along with addition of RC filter on each channel's common-mode reference (VOCM)
- Charge injection DAC linearity improved along with optional slope matching with Ch0 gain
- Soft connected V_{ss0} to V_{ss} internally to remove body biasing while improving substrate noise rejection. Hence, V_{ss0} pads (41,42,79 and 80) are now V_{ss}

LArASIC Layout and Packaging

The top-level layout with pad information and bonding diagrams are shown in Fig. 3 and Fig. 4, respectively. LArASIC has in total of 100 pads and will be packaged in the LQFP128 package. Table 3 presents layout and pad information and Table 4 presents the pin and pad list of LArASIC, respectively.

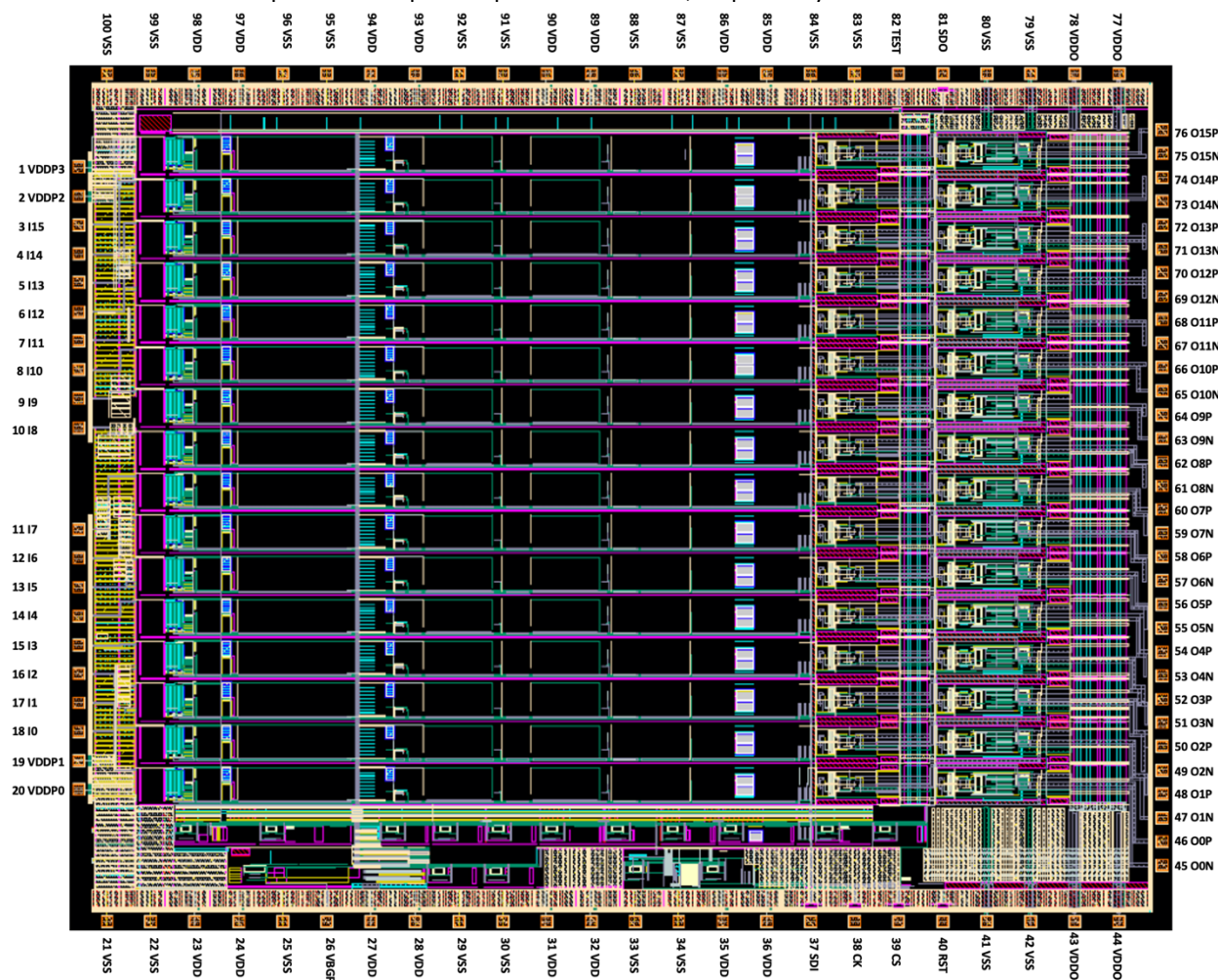


Fig. 3. LArASIC P5 layout with I/O pad names

Table 3. Layout and pad information

Layout size	7.28 x 5.7 mm ²
Die cut size	7.4 x 5.9 mm ²
Pad count	100
Pad size	78.04 x 78.04 μm ²
Pad pitch	192 μm (left side), 157.36 μm (right side) and 292.32 μm (top/bottom)

Num. Pins/ Pads	Pin Num.	Pad Num.	Signal Name	In/Out	Description
4	7, 8, 25, 26	1, 2, 19, 20	VDDP (3, 2, 1, 0)		Analog supply for the 1 st stage of the charge amplifiers: +1.8V. VDDP0 (ch0~ch3), VDDP1 (ch 4~ch7), VDDP2 (ch8~ch11), VDDP3 (ch12~ch15)
21	37, 38, 41, 45, 46, 49, 50, 57, 58, 103, 104, 107, 108, 111, 112, 115, 116, 119, 120, 123, 124	21, 22, 25, 29, 30, 33, 34, 41, 42, 79, 80, 83, 84, 87, 88, 91, 92, 95, 96, 99, 100	VSS		Analog ground: 0 V
16	39, 40, 43, 44, 47, 48, 51, 52, 109, 110, 113, 114, 117, 118, 121, 122	23, 24, 27, 28, 31, 32, 35, 36, 85, 86, 89, 90, 93, 94, 97, 98	VDD		Analog supply: +1.8 V
4	59, 60, 101, 102	43, 44, 77, 78	VDDO		Analog supply for output buffer: +1.8 V
16	9~24	3~18	Charge Inputs I (15~0)	In	DC or AC coupled charge input from the detector. ESD protected (mild)
32	65~96	45~76	O(0~15){N, P}	Out	Channel analog output, ex., O5N and O5P are channel 5 outputs. ESD protected
1	42	26	VBGR		Bandgap reference monitor. ~1.18V at room temp
1	53	37	SDI	In	CMOS level. Digital serial data input. ESD protected
1	54	38	CK	In	CMOS level. Clock for shift registers. ESD protected
1	55	39	CS	In	CMOS level. On the falling edge of CS, data is latched into the shift registers. ESD protected
1	56	40	RST	In	CMOS level. Global active low reset. ESD protected
1	105	81	SDO	Out	CMOS level. The output of the shift register. Tri-stated with CS. ESD protected
1	106	82	TEST	In	Test pulse input. DAC output, Monitor output. ESD protected
100			Total pins/pads		

Table 4. Information of pads and pins of LArASIC

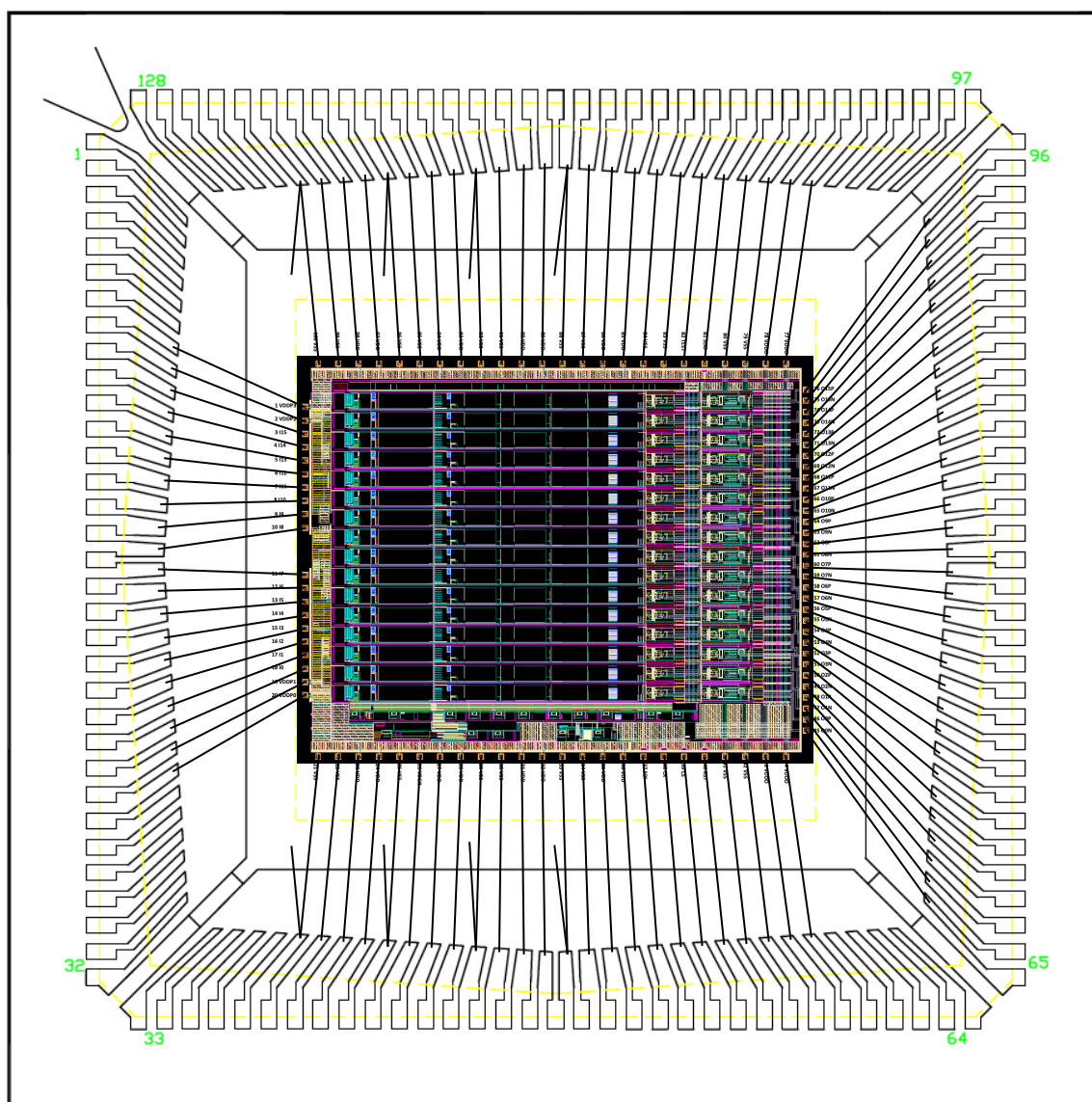


Fig. 4. Bonding diagram in the LQFP128 package

Package Name: LQFP128

Package Cavity Size: 9.5 x 9.5 mm²

Downbond-to-substrate package-pin numbers: 37, 41, 45, 49, 112, 116, 120, 124

Min. bond pad size X: 78.04 μm

Min. bond pad size Y: 78.04 μm

Min. pad pitch: 157.36 μm

Min. pad spacing: 79.32 μm

Configuration Registers

The ASIC has two 8-bit global shift registers and sixteen channel shift registers (each 8-bits long). The ASIC has four global configuration inputs: Clock (CK), Chip Select (CS), Reset (RST), and Serial Data Input (SDI); and one global configuration output: Serial Data Output (SDO). The serial data map is shown in Fig. 5.

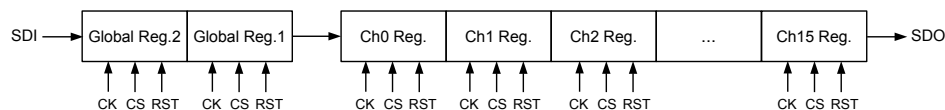


Fig. 5. Serial data map

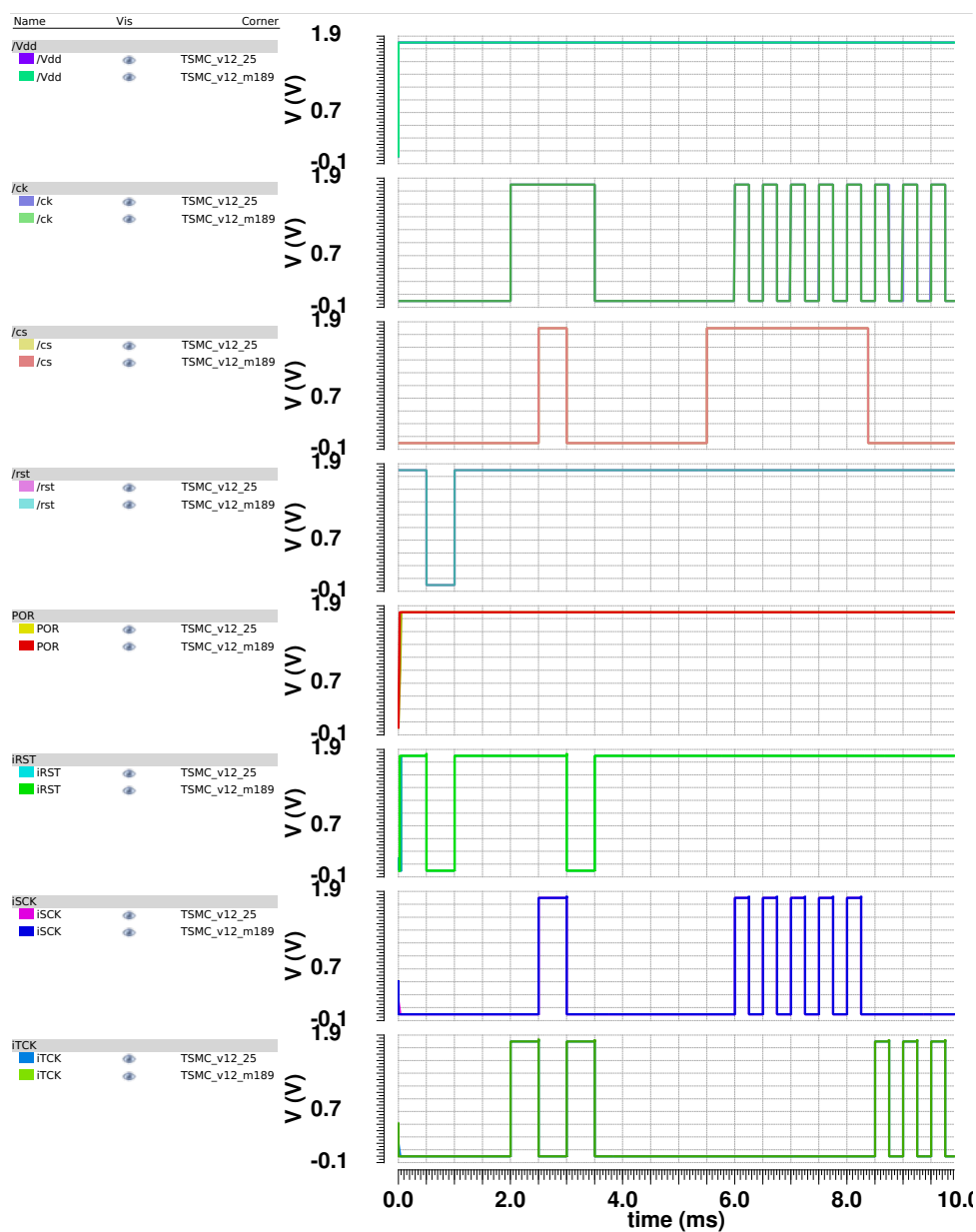


Fig. 6. A depiction of clock and other control signals along with power-on-reset (POR) and smart-reset (iRST) signals at room and cold temperatures

There are three ways to generate reset signal (iRST), whose example is also depicted in Fig. 6. :

1. Firstly, the reset (iRST) is automatically generated when the chip powers-up, known as power-on-reset (POR)
2. With the reset pin (RST), reset signal can be sent from outside
3. A Reset, known as smart-reset is also generated internally with a CS pulse if CK is kept high. The duration of the reset is from the falling-edge of CS to the falling edge of CK.

Also shown in Fig. 6 is an example of the internal signals iSCK and iTCK, which are CS-controlled configuration clock and CS-controlled test pulse clock, respectively. The test pulse clock is needed for the internal pulser as explained later in the Pulser Section.

An 8-bit shift register, and the register map is as shown in Fig. 7. As shown in Fig. 8, the 8-bit register subsystem consists of 8 D flip-flops on the bottom and 8 D flip-flops on the top, in total 16 D flip-flops are present. The outputs of first 8 D flip-flops are $s0^*$, $s1^*$, ..., $s7^*$ are available only internally. The outputs of the other 8 are $s0$, $s1$, ..., $s7$, which are the main outputs are available at the configuration pins.

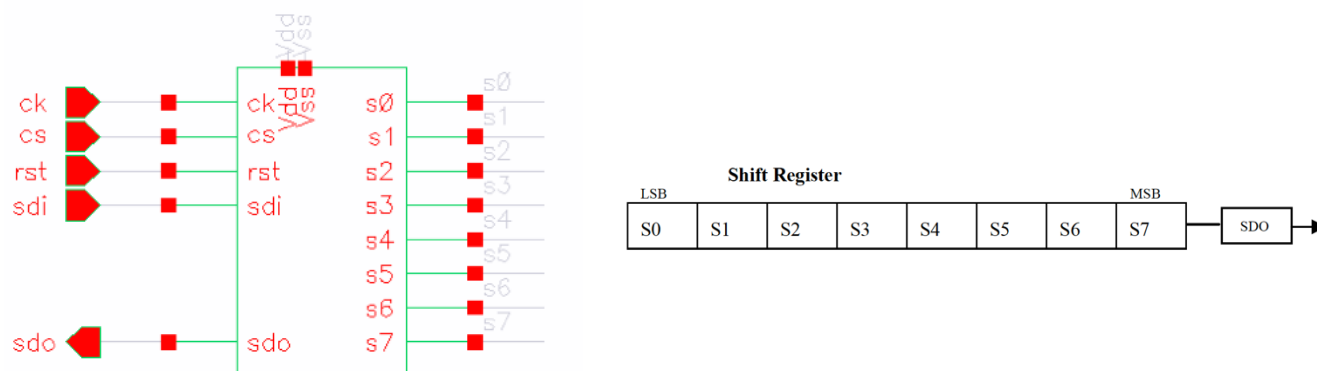


Fig. 7. An 8-bit shift register and register map

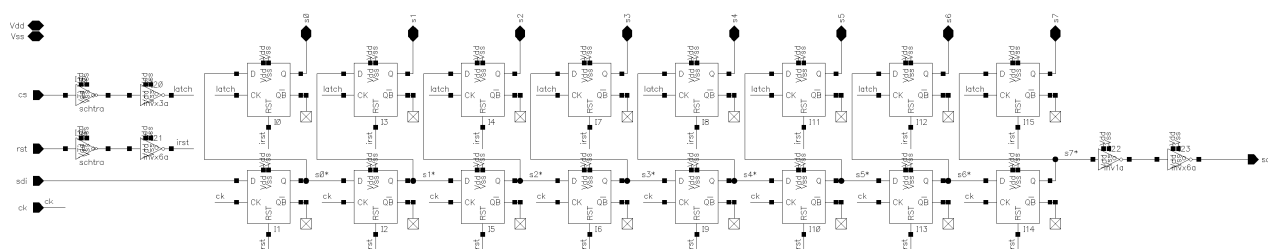


Fig. 8. Register subsystem with 16 D flip-flops

As shown in Fig. 9, at the falling edge of each CK, the configuration data (sdi) is serially shifted into the first group of 8 internal shift registers ($s0^*$, $s1^*$, ..., $s7^*$). At the falling edge of CS, the data is latched from the shift registers to the configuration pins ($s0$, $s1$, ..., $s7$). Edges on CS should not coincide with CK to setup-time violation. The output of each shift register is serially available through its SDO pin, which is a buffered version of $s7^*$.

The shift register mechanism is as explained below:

- Data is shifted into the load shift register on the rising edge of CK while CS is high.
- The MSB is shifted into position $s0^*$ on the first rising edge of CK.
- The LSB is shifted into position $s0^*$ on the 8th rising edge of CK.
- The MSB is shifted out of SDO on the 9th rising edge of CK.
- The LSB is shifted out of SDO on the 16th rising edge of CK.

NOTE: The default value of each register is 000.

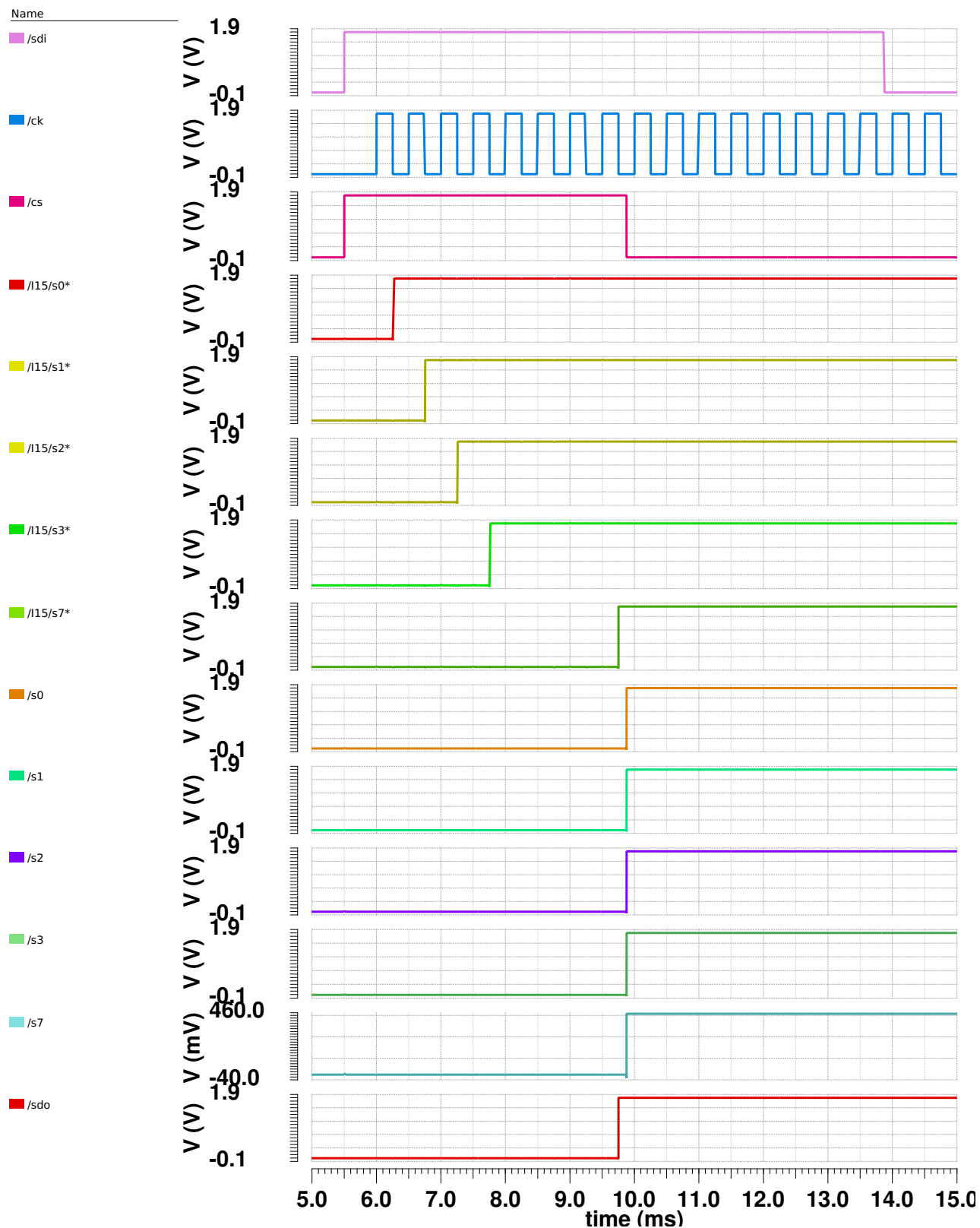


Fig. 9. A depiction of SPI configuration mechanism along with various control signals

Global Register Configuration:

The main global register depicted as Global Reg. 1 in Fig. 5 is as shown in Fig. 10. Apart from that, another 8-bit global configuration register is depicted as Global Reg. 2 in Fig. 5 is used for the DAC control and is explained later in the DAC section.

The first bit of the global register (Global Reg. 1) is unused and is reserved and hence depicted as RES. Bit STB sets Channel 0 to monitor either the analog channel signal or the temperature/bandgap reference (dedicated by bit STB1). Bit SLK sets the leakage current of each channel to either 100 pA or 500 pA. Bit SLKH increases the leakage current by a factor 10. It is summarized as shown in Table 5.



Fig. 10. The main Global register of LArASIC

Table 5. Global register description

Global Register Bit	Name	Description
0	SGP	Bit used to disable DAC gain matching with the gain setting of Ch0 (SG0, SG1) 0 – enabled 1 – disabled Explained further in the DAC section.
1	SDD	Bit used to enable SEDCs in all channels. 0 – disabled 1 – enabled Explained further in the buffer control section.
2	SDC	Output coupling. 0 – dc coupling 1 – ac coupling
3	SLKH	Reset Quiescent Current (RQI) increases by a factor of 10. 0 – disabled 1 – enabled
4	S16	Enable high filter in ch15 (16 th channel) 0 – disabled 1 – enabled
5	STB	0 – Monitor analog channel signal 1 – Monitor temperature or bandgap reference.
6	STB1	0 – Monitor temperature 1 – Monitor bandgap reference.
7	SLK	Reset Quiescent Current (RQI) setting. 0 – 500 pA RQI 1 – 100 pA RQI

Channel Register Configuration:

Each channel register has 8 bits: STS, SNC, SG0, SG1, ST0, ST1, SDC, and SDF as depicted in Fig. 11.

- STS enables the 200 fF test capacitor individually for each channel. The analog test pulse can be applied through pin TEST. The pin can be terminated (50 Ohm) when not connected to a pulse generator.
- SNC selects the baseline to either 200 mV (for a unipolar pulse in collecting mode) or 900 mV (for a bipolar pulse in induction/non-collecting mode).
- The channel gain can be independently adjusted to 4.7, 7.8, 14, or 25 mV/fC, through two dedicated bits (SG0 and SG1).
- The peaking time of each channel can be set independently to 0.5, 1, 2, or 3 μ s through two dedicated bits (ST0 and ST1).
- SDF sets the output buffer to either selected or bypassed (also powered down).

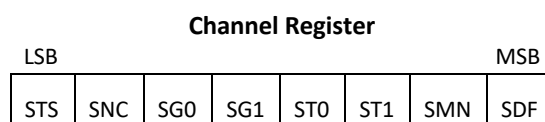


Fig. 11. Channel register present in each of the LArASIC channel

Table 6. summarizes channel register settings.

Table 6. Channel register description and possible configurations

Channel Register Bit	Name	Description
0	STS	Test capacitance. 0 – disabled 1 – enabled
1	SNC	Baseline selection. 0 – 900 mV (for non-collecting mode) 1 – 200 mV (for collecting mode).
2:3	SG (0,1)	Gain selection. 00 – 14 mV/fC 10 – 25 mV/fC 01 – 7.8 mV/fC 11 – 4.7 mV/fC
4:5	ST (0,1)	Peak time selection. 00 – 1.0 μ s 10 – 0.5 μ s 01 – 3 μ s 11 – 2 μ s
6	SMN	Output monitor enable. 0 – monitor disabled 1 – monitor enabled: channel output routed to Test pad. Conflict configurations exist and are discussed in the DAC section.
7	SDF	Explained in the next section (Buffer Control Configuration)

Buffer Control Configuration:

With the addition of single-ended-to-differential conversion (SEDC) buffer (starting with LArASIC P4) in addition to the pre-existing single-ended (SE) buffer, the buffer control configuration mechanism is as depicted in Fig. 12. and the corresponding three possible scenarios are shown in Table 7.

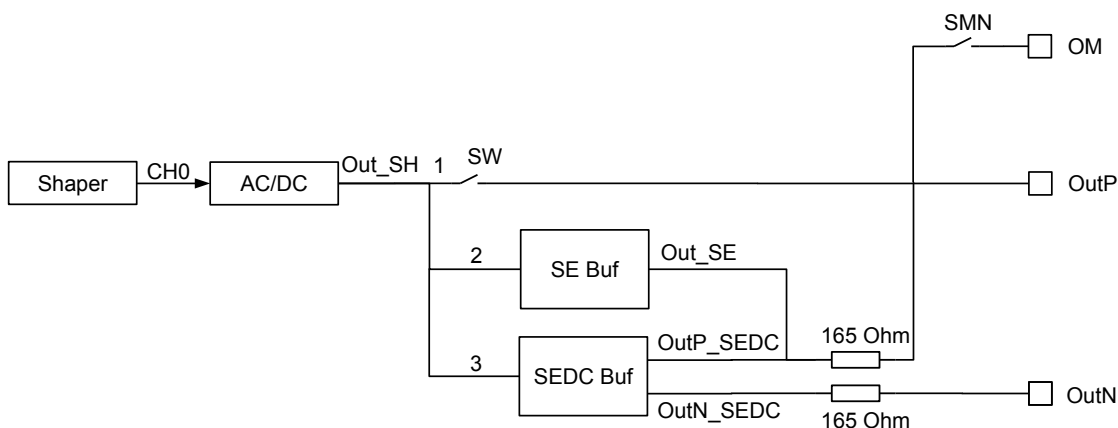


Fig. 12. Depiction of buffer control configuration

Table 7. Buffer control configuration for the three different scenarios

Scenario	SDD (Global reg.)	SDF (Channel Reg.)	SW	Power Down	OutP	OutN
1	0	0	Close	SE & SEDC	Out_SH	High Z
2	0	1	Open	SEDC	Out_SE	High Z
3	1	X	Open	SE	OutP_SEDC	OutN_SEDC

On-Chip DAC and Pulse Generator

The ASIC contains a 6-bit Current Scaling DAC and a pulse generator for the test and calibration of the front-end channels.

In LArASIC P5 the DAC gain is matched to the gain setting of Ch0 (SG0, SG1), which can be overridden with SGP global register bit as shown in Table 8.

Table 8. Pulser DAC programmability

SGP (Global bit)	Channel 0 Register		Gain Selection (mV/fC)
	SG0	SG1	
0	0	0	14
	1	0	25
	0	1	7.8
	1	1	4.7
1	X	X	4.7

The specifications of the DAC are as shown in Table 9 and as follows:

- Power: 1 mW
- Temperature Range: 27 °C to -200 °C
- Settling time: < 130 ns
- Target Linearity: < $\pm 0.1\%$

Table 9. Pulser DAC specifications for the four gain settings

Full-Scale Input Charge (fC)	300	180	100	56
Gain each stage (mV/fC)	4.7	7.8	14	25
Injected Charge	238	180	100	56
Peak to Peak Voltage	1.2	0.9	0.5	0.28
R_{FS} (k Ω)	10	7.68	4.27	2.39

Where R_{FS} is the resistor used to program the gain.

Global register 2 is added before the first one to provide the inputs for the DAC as depicted in Fig. 13

Global Register 2

LSB				MSB			
SDAC0	SDAC1	SDAC2	SDAC3	SDAC4	SDAC5	SDACSW1	SDACSW2

Fig. 13. Global register 2 used for DAC control

- sdac0 ~ sdac5:** 6-bit input to the DAC where sdac0 is the LSB and sdac5 is the MSB
- sdacsw1 :** 0 – test input to the channels is disconnected from the external test pin.
1 – test input to the channels is connected to the external test pin.
- sdacsw2:** 0 – test input to the channels is disconnected from the DAC output.
1 – test input to the channels is connected to the DAC output.

Do not use SMN with SDACSW1 and/or SDACSW2 high. Do not enable more than one monitor at a time. The following are the conflict configurations:

1. SMN+SDACSW1 – will short the channel output with the injection input
2. SMN+SDACSW1+SDACSW2 – will short the channel output with the DAC output

The output of the DAC is fed to a pulse generator that is controlled by CK clock input to generate the output voltage pulse. Rising edge of clock injects positive charge while falling edge injects negative. The scheme is as shown in Fig. 14.

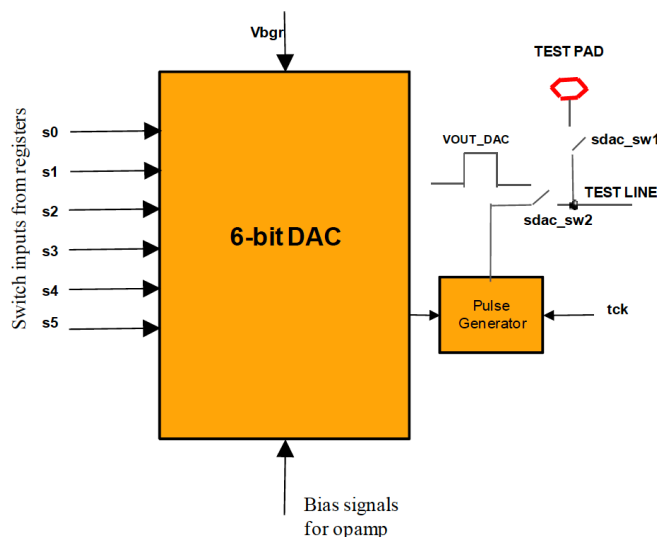


Fig. 14. On-chip DAC and pulse generator

Fig. 15 shows typical DAC output characteristic at room and cold temperatures.

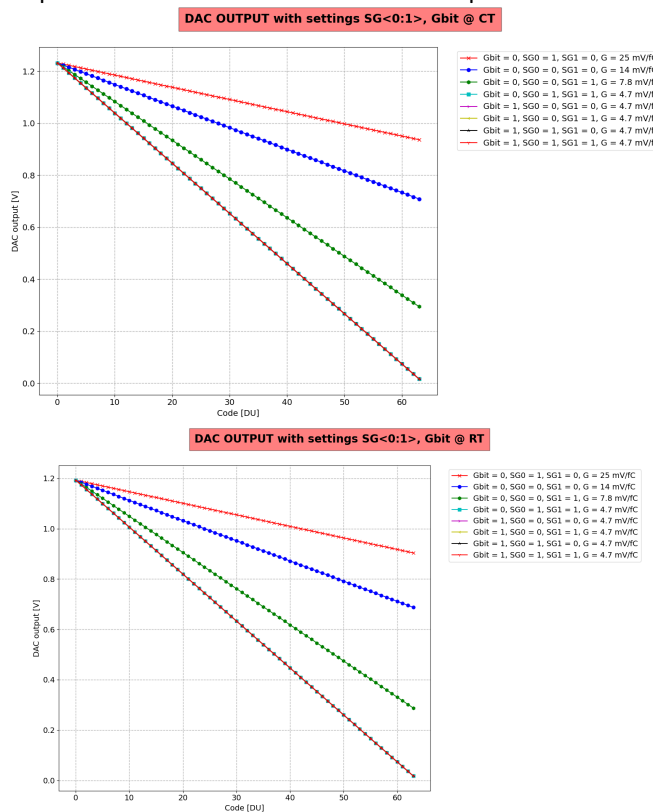


Fig. 15. DAC output characteristic at room and cold temperatures

Fig. 16 shows typical waveforms that can be expected at the outputs of the channels.

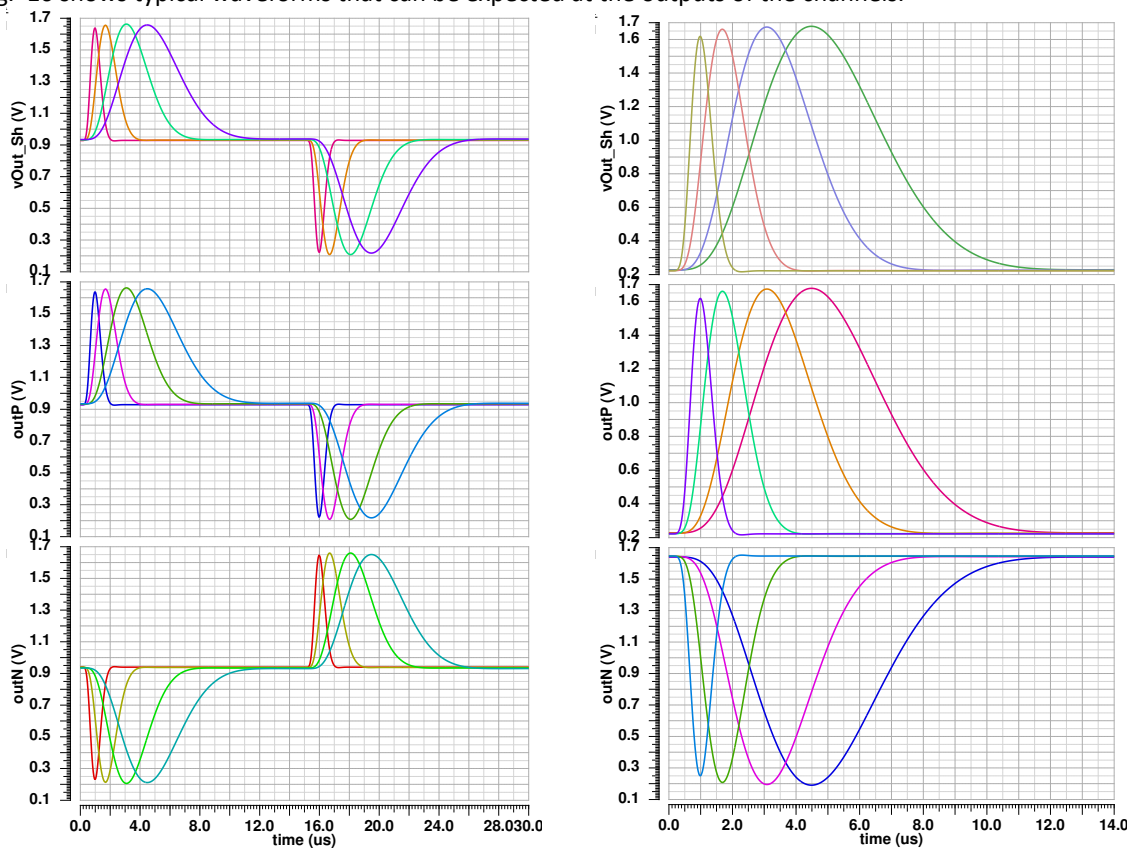


Fig. 16 Simulated outputs (shaper, outP, outN) for all the four peaking time settings in induction mode (left) with 900 mV baseline, and collection mode (right) with 200 mV baseline. Load = 20 pF || 250 kΩ, temperature = 77K, Cdet = 150 pF. Channel settings: RQI = 500 pA (SLK = 0, SLKH = 0), SEDC enabled (SDD = 1), gain = 14 mV/fC (SG = 00), DC coupling (SDC = 0).

Fig. 17 shows the expected ENC.

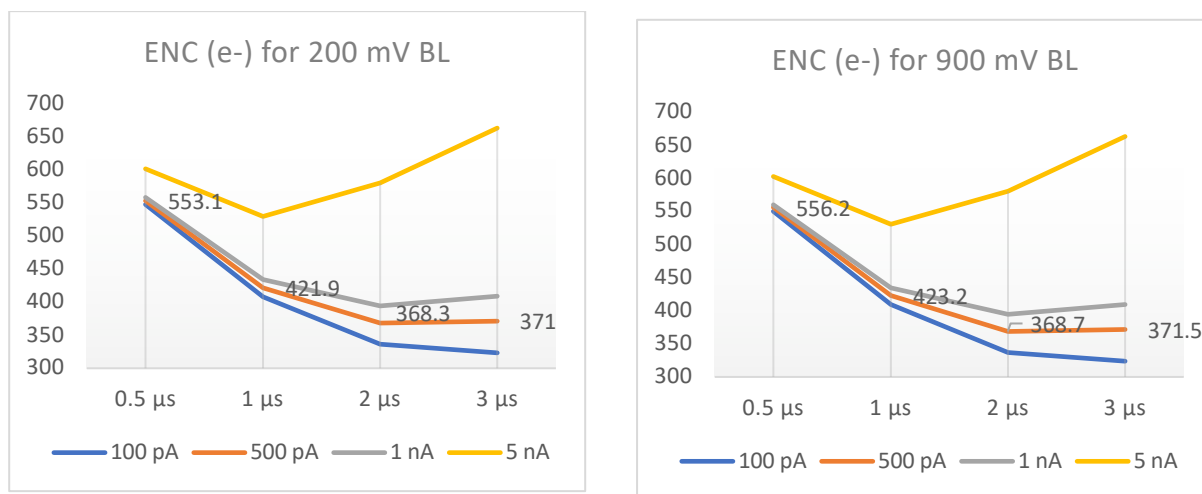


Fig. 17 Simulated ENC of the channel without SE or SEDC Buffers. The ENC with SE or SEDC stays same. Load = 20 pF || 250 kΩ, temperature = 77K, Cdet = 150 pF. Channel settings: gain = 14 mV/fC (SG = 00), DC coupling (SDC = 0).

Simulations of the channel linearity are presented in Fig. 18 and Fig. 19 for the channel with SE buffer and SEDC buffer, respectively. Triangular charge input pulses mimicking the physics signals are used for the simulation, example input and output plots are shown in Fig. 20.

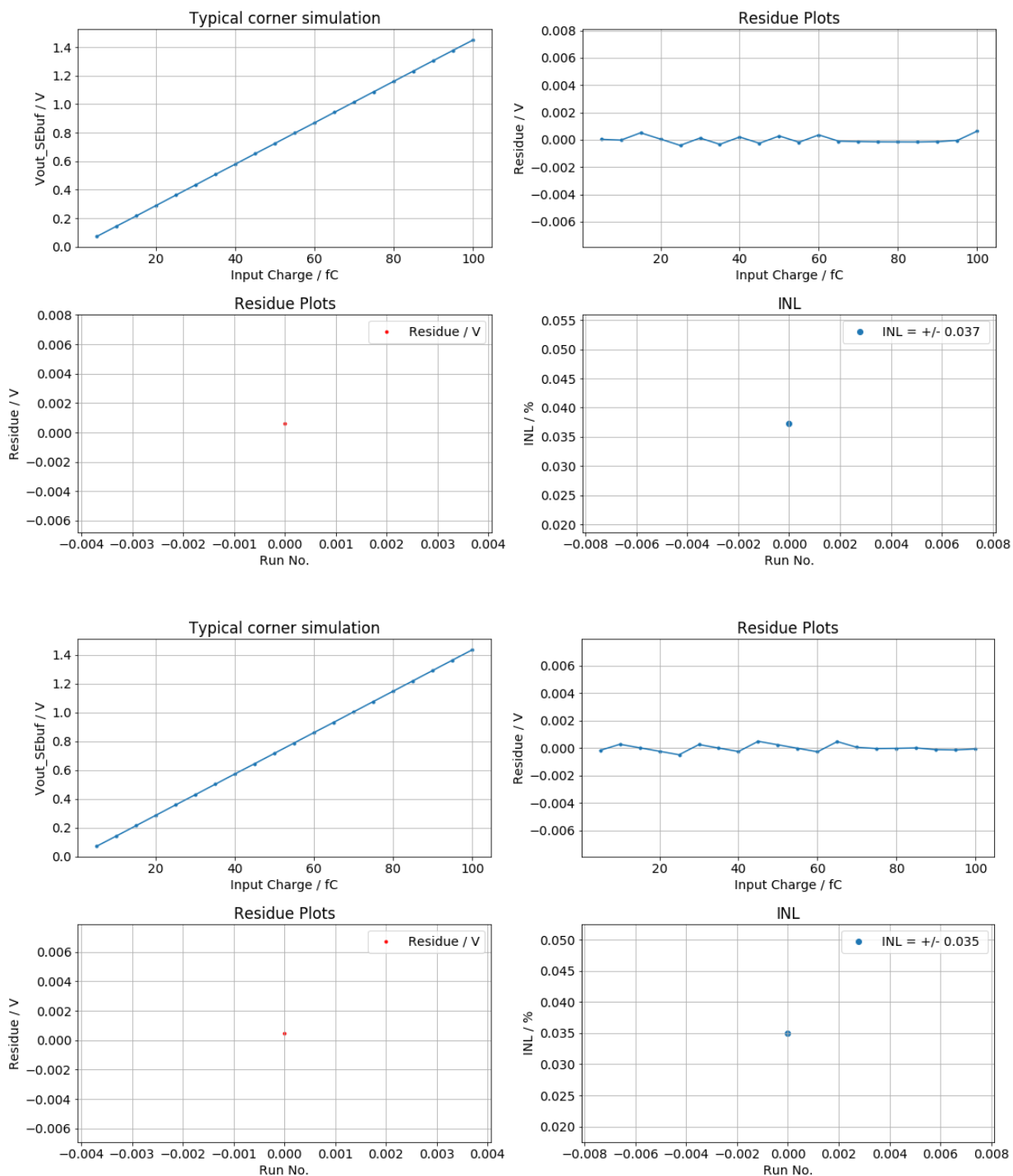


Fig. 18 Typical corner simulation of dynamic nonlinearity of channel with SE buffer for the peaking time of $2\mu s$ at $25^\circ C$ with BSIM4.5 models (top) and $-189^\circ C$ with PSP models (bottom). Load = $20 pF \parallel 250 k\Omega$, $C_{det} = 150 pF$. Channel settings: $RQ1 = 500 pA$ ($SLK = 0$, $SLKH = 0$), SE enabled ($SDD = 0$, $SDF = 1$), gain = $14 mV/fC$ ($SG = 00$), DC coupling ($SDC = 0$)

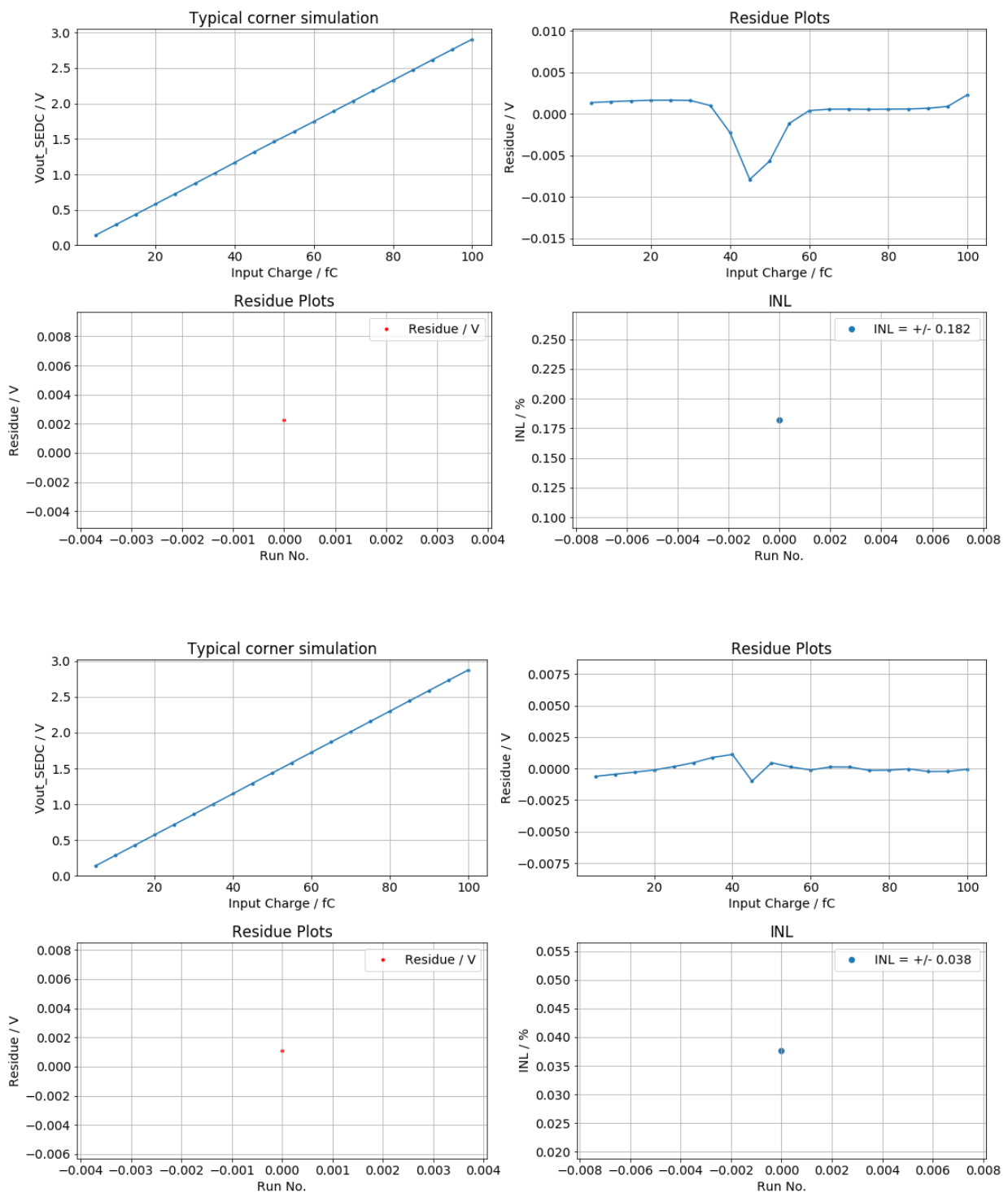


Fig. 19 Typical corner simulation of dynamic nonlinearity of channel with SEDC buffer for the peaking time of $2\mu s$ at 25 °C with BSIM4.5 models (top) and -189 °C with PSP models (bottom). Load = 20 pF || 250 k Ω , Cdet = 150 pF. Channel settings: RQ1 = 500 pA (SLK = 0, SLKH = 0), SE enabled (SDD = 1, SDF = X), gain = 14 mV/fC (SG = 00), DC coupling (SDC = 0).

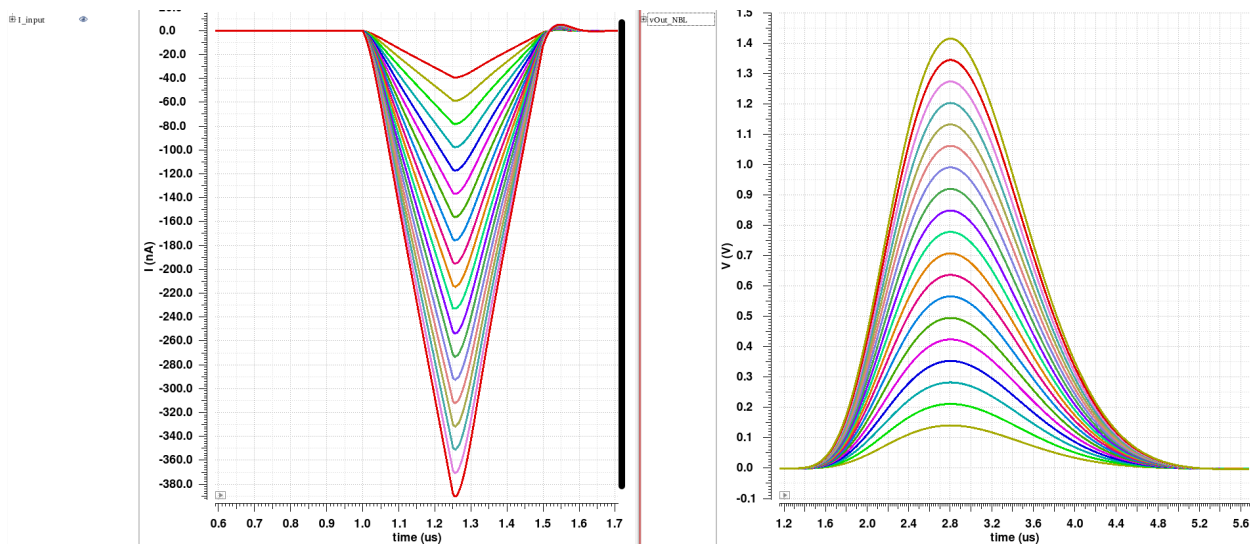


Fig. 20 Example of input and output signals for measuring the linearity of the FE

Appendix A: Pin and Pad Information

Table 10 Left side package pins and silicon pads information

Pad Number	Pad Name	X Center (μm)	Y Center (μm)	Pin Number	Pin Name	Comments
1	VDDP3	39.02	5047.665	7	VDDP3	
2	VDDP2	39.02	4855.665	8	VDDP2	
3	I15	39.02	4663.665	9	I15	
4	I14	39.02	4471.665	10	I14	
5	I13	39.02	4279.665	11	I13	
6	I12	39.02	4087.665	12	I12	
7	I11	39.02	3895.665	13	I11	
8	I10	39.02	3703.665	14	I10	
9	I9	39.02	3511.665	15	I9	
10	I8	39.02	3319.665	16	I8	
11	I7	39.02	2643.715	17	I7	
12	I6	39.02	2451.715	18	I6	
13	I5	39.02	2259.715	19	I5	
14	I4	39.02	2067.715	20	I4	
15	I3	39.02	1875.715	21	I3	
16	I2	39.02	1683.715	22	I2	
17	I1	39.02	1491.715	23	I1	
18	I0	39.02	1299.715	24	I0	
19	VDDP1	39.02	1107.715	25	VDDP1	
20	VDDP0	39.02	915.715	26	VDDP0	

Table 11 Bottom side package pins and silicon pads information

Pad Number	Pad Name	X Center (μm)	Y Center (μm)	Pin Number	Pin Name	Comments
21	VSS	227.445	39.02	37	VSS	Downbond-to-substrate
22	VSS	519.765	39.02	38	VSS	
23	VDD	812.085	39.02	39	VDD	
24	VDD	1104.405	39.02	40	VDD	
25	VSS	1396.725	39.02	41	VSS	Downbond-to-substrate
26	VBGR	1689.045	39.02	42	VBGR	
27	VDD	1981.365	39.02	43	VDD	
28	VDD	2273.685	39.02	44	VDD	
29	VSS	2566.005	39.02	45	VSS	Downbond-to-substrate
30	VSS	2858.325	39.02	46	VSS	
31	VDD	3150.645	39.02	47	VDD	
32	VDD	3442.965	39.02	48	VDD	
33	VSS	3735.285	39.02	49	VSS	Downbond-to-substrate
34	VSS	4027.605	39.02	50	VSS	
35	VDD	4319.925	39.02	51	VSS	
36	VDD	4612.245	39.02	52	VDD	
37	SDI	4904.565	39.02	53	SDI	
38	CK	5196.885	39.02	54	CK	
39	CS	5489.205	39.02	55	CS	
40	RST	5781.525	39.02	56	RST	
41	VSS	6073.845	39.02	57	VSS	
42	VSS	6366.165	39.02	58	VSS	
43	VDDO	6658.485	39.02	59	VDDO	
44	VDDO	6950.805	39.02	60	VDDO	

Table 12 Right side package pins and silicon pads information

Pad Number	Pad Name	X Center (μm)	Y Center (μm)	Pin Number	Pin Name	Comments
45	O0N	7240.11	414.5	65	O0N	
46	O0P	7240.11	571.86	66	O0P	
47	O1N	7240.11	729.22	67	O1N	
48	O1P	7240.11	886.58	68	O1P	
49	O2N	7240.11	1043.94	69	O2N	
50	O2P	7240.11	1201.3	70	O2P	
51	O3N	7240.11	1358.66	71	O3N	
52	O3P	7240.11	1516.02	72	O3P	
53	O4N	7240.11	1673.38	73	O4N	
54	O4P	7240.11	1830.74	74	O4P	
55	O5N	7240.11	1988.1	75	O5N	
56	O5P	7240.11	2145.46	76	O5P	
57	O6N	7240.11	2302.82	77	O6N	
58	O6P	7240.11	2460.18	78	O6P	
59	O7N	7240.11	2617.54	79	O7N	
60	O7P	7240.11	2774.9	80	O7P	
61	O8N	7240.11	2932.26	81	O8N	
62	O8P	7240.11	3089.62	82	O8P	
63	O9N	7240.11	3246.98	83	O9N	
64	O9P	7240.11	3404.34	84	O9P	
65	O10N	7240.11	3561.7	85	O10N	
66	O10P	7240.11	3719.06	86	O10P	
67	O11N	7240.11	3876.42	87	O11N	
68	O11P	7240.11	4033.78	88	O11P	
69	O12N	7240.11	4191.14	89	O12N	
70	O12P	7240.11	4348.5	90	O12P	
71	O13N	7240.11	4505.86	91	O13N	
72	O13P	7240.11	4663.22	92	O13P	
73	O14N	7240.11	4820.58	93	O14N	
74	O14P	7240.11	4977.94	94	O14P	
75	O15N	7240.11	5135.3	95	O15N	
76	O15P	7240.11	5292.66	96	O15P	

Table 13 Topside package pins and silicon pads information

Pad Number	Pad Name	X Center (μm)	Y Center (μm)	Pin Number	Pin Name	Comments
77	VDDO	6950.805	5668.16	101	VDDO	
78	VDDO	6658.485	5668.16	102	VDDO	
79	VSS	6366.165	5668.16	103	VSS	
80	VSS	6073.845	5668.16	104	VSS	
81	SDO	5781.525	5668.16	105	SDO	
82	TEST	5489.205	5668.16	106	TEST	
83	VSS	5196.885	5668.16	107	VSS	
84	VSS	4904.565	5668.16	108	VSS	
85	VDD	4612.245	5668.16	109	VDD	
86	VDD	4319.925	5668.16	110	VDD	
87	VSS	4027.605	5668.16	111	VSS	
88	VSS	3735.285	5668.16	112	VSS	Downbond-to-substrate
89	VDD	3442.965	5668.16	113	VDD	
90	VDD	3150.645	5668.16	114	VDD	
91	VSS	2858.325	5668.16	115	VSS	
92	VSS	2566.005	5668.16	116	VSS	Downbond-to-substrate
93	VDD	2273.685	5668.16	117	VDD	
94	VDD	1981.365	5668.16	118	VDD	
95	VSS	1689.045	5668.16	119	VSS	
96	VSS	1396.725	5668.16	120	VSS	Downbond-to-substrate
97	VDD	1104.405	5668.16	121	VDD	
98	VDD	812.085	5668.16	122	VDD	
99	VSS	519.765	5668.16	123	VSS	
100	VSS	227.445	5668.16	124	VSS	Downbond-to-substrate